

We conclude that $\Delta = 1$, and hence Γ^c has precisely two components. Finally, only one of these components can be unbounded, namely $\mathcal{U}$, and the winding number of Γ in this component must therefore be zero. By our last result, we see that, after possibly reversing the orientation of the curve, the winding number of any point in the bounded component Ω is constant and equal to 1. Also, it is clear from what has been said that any smooth point on Γ can be approached by points in either component, and hence Γ is the boundary of both Ω and $\mathcal{U}$.

The final step in the proof is to show that the interior of the curve, that is, the bounded component Ω , is simply connected. By Theorem 1.2 it suffices to show that Ω^c is connected. If not, then

$$\Omega^c = F_1 \cup F_2,$$

where F_1 and F_2 are closed, disjoint, and non-empty. Let

$$\mathcal{O}_1 = \mathcal{U} \cap F_1 \quad \text{and} \quad \mathcal{O}_2 = \mathcal{U} \cap F_2.$$

Clearly, $\mathcal{O}_1$ and $\mathcal{O}_2$ are disjoint. If $z \in \mathcal{O}_1$, then $z \in \mathcal{U}$, and every small ball centered at z is contained in $\mathcal{U}$. If every such ball intersects F_2 , then $z \in F_2$ since F_2 is closed. However, F_1 and F_2 are disjoint, so this cannot happen. Consequently, $\mathcal{O}_1$ is open, and by the same argument, so is $\mathcal{O}_2$. Finally, we claim that $\mathcal{O}_1$ is non-empty. If not, then F_1 is entirely contained in Γ and $\mathcal{U}$ is contained in F_2 . Pick any point $z \in F_1$, which we know belongs to Γ . Now every ball centered at z intersects $\mathcal{U}$, hence F_2 . But F_2 is closed and disjoint from F_1 , so we get a contradiction. A similar argument for $\mathcal{O}_1$ proves that

$$\mathcal{U} = \mathcal{O}_1 \cup \mathcal{O}_2,$$

where $\mathcal{O}_1, \mathcal{O}_2$ are disjoint, open, and non-empty. This contradicts the fact that $\mathcal{U}$ is connected, and concludes the proof of the Jordan curve theorem for piecewise-smooth curves.

2.1 Proof of a general form of Cauchy's theorem

Theorem 2.9 *If a function f is holomorphic in an open set that contains a simple closed piecewise-smooth curve Γ and its interior, then*

$$\int_{\Gamma} f = 0.$$

Let $\mathcal{O}$ denote an open set on which f is holomorphic, and which contains Γ and its interior Ω . The idea is to construct a closed curve Λ

in Ω that is so close to Γ that $\int_{\Gamma} f = \int_{\Lambda} f$. Then, the integral on the right-hand side is 0, since f is holomorphic in the simply connected open set Ω . We build Λ as follows. Near the smooth parts of Γ , the curve Λ is essentially a curve like Γ_{ϵ} in Lemma 2.4. Near points where smooth parts of Γ join, we shall use for Λ an arc of a circle. This is illustrated in Figure 5.

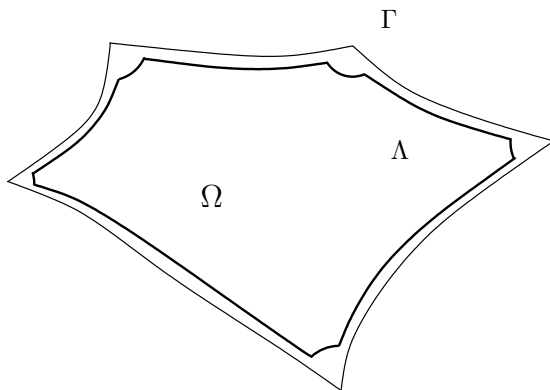


Figure 5. The curve Λ

To find the appropriate connecting arcs, we need the following preliminary result.

Lemma 2.10 *Let $\gamma : [0, 1] \rightarrow \mathbb{C}$ be a simple smooth curve. Then, for all sufficiently small $\delta > 0$ the circle C_{δ} centered at $\gamma(0)$ and of radius δ intersects γ in precisely one point.*

Proof. We may assume that $\gamma(0) = 0$. Since $\gamma(0) \neq \gamma(1)$ it is clear that for each small $\delta > 0$, the circle C_{δ} intersects γ in at least one point. If the conclusion in the lemma is false, we can find a sequence of positive δ_j going to 0, and so that the equation $|\gamma(t)| = \delta_j$ has at least two distinct solutions. The mean value theorem applied to $h(t) = |\gamma(t)|^2$ provides a sequence of positive numbers t_j so that $t_j \rightarrow 0$ and $h'(t_j) = 0$. Thus

$$\gamma'(t_j) \cdot \gamma(t_j) = 0 \quad \text{for all } j.$$

However, the curve is smooth, so

$$\gamma(t) = \gamma(0) + \gamma'(0)t + t\varphi(t) \quad \text{and} \quad \gamma'(t) = \gamma'(0) + \psi(t),$$

where $|\varphi(t)| \rightarrow 0$ and $|\psi(t)| \rightarrow 0$ as t goes to 0. Then recalling that $\gamma(0) = 0$, we find $\gamma'(t) \cdot \gamma(t) = |\gamma'(0)|^2 t + o(|t|)$. The definition of a smooth curve also requires that $\gamma'(0) \neq 0$, so the above gives

$$\gamma'(t) \cdot \gamma(t) \neq 0 \quad \text{for all small } t.$$

This is the desired contradiction.

Returning to the proof of Cauchy's theorem, choose ϵ so small that the open set $\mathcal{U}$ of all points at a distance $< \epsilon$ of Γ is contained in $\mathcal{O}$.

Next, if $P_1, \dots, P_n$ denote the consecutive points where smooth parts of Γ join, we may pick $\delta < \epsilon/10$ so small that each circle C_j centered at a point P_j and of radius δ intersects Γ in precisely two distinct points (this is possible by the previous lemma). These two points on C_j determine two arcs of circles, only one of which (denoted by C_j) has an interior entirely contained in Ω . To see this, it suffices to recall that if γ is a parametrization of a smooth part of Γ with end-point P_j , then for all small ϵ' the curves parametrized by $\gamma_{\epsilon'}$ and $\gamma_{-\epsilon'}$ of Lemma 2.4 lie on opposite sides of Γ and must intersect the circle C_j . By construction the disc D_j^* centered at P_j and of radius 2δ is also contained in $\mathcal{U}$, hence in $\mathcal{O}$.

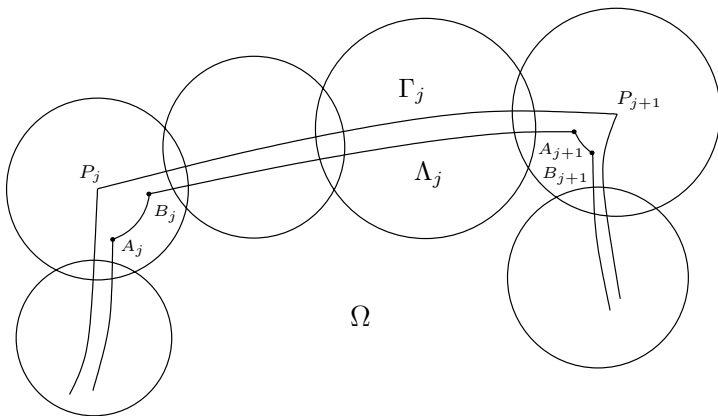


Figure 6. Construction of the curve Λ

We wish to construct Λ so that we may argue as in the proof of Theorem 5.1, Chapter 3 and establish $\int_{\Gamma} f = \int_{\Lambda} f$. To do so, we consider a chain of discs $\mathcal{D} = \{D_0, \dots, D_K\}$ contained in $\mathcal{U}$, and so that Γ is contained in their union, with $D_k \cap D_{k+1} \neq \emptyset$, $D_0 = D_K$, and with the discs

D_j^* part of the chain $\mathcal{D}$. Suppose Γ_j is the smooth part of Γ that joins P_j to P_{j+1} . By Lemma 2.4 it is possible to construct a continuous curve Λ_j that is contained in Ω and in the union of the discs, and which connects a point on B_j on $\mathcal{C}_j$ to a point on A_{j+1} on $\mathcal{C}_{j+1}$ (see Figure 6). Since we only assumed that Γ has one continuous derivative, Λ_j need not be smooth, but by approximating this continuous curve by polygonal lines if necessary, we may actually assume that Λ_j is also smooth. Then, A_{j+1} is joined to B_{j+1} by a piece of $\mathcal{C}_{j+1}$, and so on. This procedure provides a piecewise-smooth curve Λ that is closed and contained in Ω .

Since f has a primitive on each disc of the family $\mathcal{D}$, we may argue as in the proof of Theorem 5.1, Chapter 3 to find that $\int_{\Gamma} f = \int_{\Lambda} f$. Since Ω is simply connected, we have $\int_{\Lambda} f = 0$, and as a result

$$\int_{\Gamma} f = 0.$$

Notes and References

Useful references for many of the subjects treated here are Saks and Zygmund [34], Ahlfors [2], and Lang [23].

Introduction

The citation is from Riemann's dissertation [32].

Chapter 1

The citation is a free translation of a passage in Borel's book [6].

Chapter 2

The citation is a translation of an excerpt from Cauchy's memoir [7].

Results related to the natural boundaries of holomorphic functions in the unit disc can be found in Titchmarsh [36].

The construction of the universal functions in Problem 5 are due to G. D. Birkhoff and G.R. MacLane.

Chapter 3

The citation is a translation of a passage in Cauchy's memoir [8].

Problem 1 and other results related to injective holomorphic mappings (univalent functions) can be found in Duren [11].

Also, see Muskhelishvili [25] for more about the Cauchy integral introduced in Problem 5.

Chapter 4

The citation is from Wiener [40].

The argument in Exercise 1 was discovered by D. J. Newman; see [4].

The Paley-Wiener theorems appeared first in [28]; further generalizations can be found in Stein and Weiss [35].

Results related to the Borel transform (Problem 4) can be found in Boas [5].

Chapter 5

The citation is a translation from the German of a passage in a letter from K. Weierstrass to S. Kowalewskaja; see [38].

A classical reference for Nevanlinna theory is the book by R. Nevanlinna himself [27].

Chapter 6

A number of different proofs of the analytic continuation and functional equation for the zeta function can be found in Chapter 2 of Titchmarsh [37].

Chapter 7

The citation is from Hadamard [14]. Riemann's statement concerning the zeroes of the zeta function in the critical strip is a passage taken from his paper [33].

Further material related to the proof of the prime number theorem presented in the text is in Chapter 2 of Ingham [19], and Chapter 3 of Titchmarsh [37].